

Project Document: 100 Drone Coordination Simulation

Project Overview

This project involves simulating coordinated behavior among 100 autonomous drones in a shared simulated environment. The goal is to design and implement a **decentralized coordination system** enabling multiple drones to perform complex tasks through **local interactions**, avoiding centralized control bottlenecks. Students will develop key algorithmic components supporting scalable teamwork, including **path planning, collision avoidance, task allocation, and dynamic reconfiguration**.

Simulation will use **discrete time steps** in a **2D environment**, with simplified drone kinematics and local sensing to ensure feasible computation and emergent behavior demonstration.

Core Implementation Structure

The core implementation is divided into three milestones, each building toward demonstrating scalable, autonomous drone teamwork.

Milestone 1: Environmental Modeling and Drone Behavior Foundation

This milestone establishes the simulated environment and foundational drone control logic.

Key Activities:

- **Simulated Environment:**
 - Use a **2D continuous plane or grid-based terrain** with obstacles (static or dynamic).
 - Define boundaries and obstacle locations.
- **Drone Representation:**
 - Each drone maintains **position, velocity, and orientation**.
 - Simulate simplified kinematics (no aerodynamics) using discrete time updates: $[x_{t+1} = x_t + v \cdot \Delta t, \quad y_{t+1} = y_t + v \cdot \Delta t]$
- **Local Sensing and Interaction:**
 - Each drone detects **neighbors within a fixed interaction radius**.
 - Measure relative distance, velocity, and heading for local decision-making.
- **Collision Avoidance:**
 - Implement **reactive avoidance** rules: if two drones approach within a safety distance, adjust direction or speed.

Deliverables:

- Discrete 2D simulation environment with obstacles.
- Basic drone movement and sensing mechanics.

- Visualization of individual drone motion and neighbor awareness.
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Milestone 2: Decentralized Coordination Algorithms

This milestone focuses on distributed algorithms enabling **collective behavior without central supervision**.

Key Activities:

- **Flocking and Formation Control:**
 - Implement **alignment, cohesion, and separation** rules (Reynolds-style) for group movement.
- **Decentralized Task Allocation:**
 - Assign sub-tasks dynamically based on **proximity, capability, or priority**.
 - Use local communication only (neighbors within interaction radius).
- **Consensus and Priority Mechanisms:**
 - Implement lightweight **message-passing protocols** among nearby drones.
 - Enable drones to agree on movement direction or task priorities without global supervision.
- **Collision Avoidance Enhancements:**
 - Integrate avoidance with alignment and task decisions.

Deliverables:

- Emergent coordinated motion (flocking or formation).
- Dynamic, decentralized task allocation among drones.
- Local communication protocols for shared intent formation.

Simulation Tips:

- Use **Python + Pygame, Matplotlib, or Unity (2D)** for visualization.
 - Use **spatial partitioning** (grid, quadtree) to reduce neighbor-check complexity.
 - Limit updates to discrete timesteps (e.g., 10–50 ms per tick) for computational efficiency.
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Milestone 3: Task Execution and Team Synergy Validation

The final milestone integrates all components to demonstrate **emergent teamwork** in a multi-phase mission.

Key Activities:

- **Mission Execution:**
 - Example missions: area coverage, target tracking, object transport, or formation traversal.
 - Each drone adapts its local behavior based on task objectives and neighbor state.
- **Feedback Loops and Adaptation:**
 - Implement **real-time adjustments** if drones fail or obstacles move dynamically.

- **Performance Evaluation:**

- Metrics include **mission completion time, collision count, area coverage**, and robustness to drone failures.

- **Visualization and Logging:**

- Provide visual demonstration of emergent behavior.
- Log metrics for quantitative analysis.

Deliverables:

- Fully integrated simulation with 100 drones performing coordinated tasks.
- Performance analysis showing efficiency, collision avoidance, and task completion.
- Demonstration of emergent behavior and robustness under disruptions.

Additional Notes / Best Practices

- **Simplified Simulation:**

- Use **2D environment**, discrete time updates, and simplified kinematics.
- Reduce computational load with **neighbor radius limitation** and **spatial partitioning**.

- **Algorithmic Design:**

- Start with basic flocking + reactive avoidance.
- Add decentralized task allocation and consensus later.

- **Reproducibility:**

- Use **fixed random seeds** to ensure consistent simulation runs.

- **Evaluation Metrics:**

- Time to complete mission
- Number of collisions
- Coverage accuracy or target tracking error
- Robustness to failures (e.g., dropped drones)

Summary: This project demonstrates scalable **autonomous drone teamwork** in a decentralized simulation. By combining simplified drone physics, local sensing, decentralized coordination, and task execution, students will learn **multi-agent systems, emergent behavior, and distributed algorithm design** while keeping simulation computationally feasible for 100 agents.